AMAM Note

1 Notation & Perturbation Method

Big $\mathcal{O}$: $f(z)=\mathcal{O}(g(z))$ as $z\to z_0$ if $\lim_{z\to z_0}\frac{f(z)}{g(z)}$ **bdd** $\Leftrightarrow |f(z)|\le A|g(z)|$ as $z\to z_0$

Small o: $f(z)=o(g(z))$ as $z\to z_0$ if $\lim_{z\to z_0}\frac{f(z)}{g(z)}=0 \Leftrightarrow f(z)\ll g(z)$ as $z\to z_0$

Asymptotic $\sim$: $f(z)\sim g(z)$ as $z\to z_0$ if $\lim_{z\to z_0}\frac{f(z)}{g(z)}=1$ **ps:** $e^{-x}=o(x^{-n}), \forall n>0$ **as** $x\to\infty$

Asymptotic Sequence: Seq $\{\varphi_n(z)\}$ is asy. if $\forall n\in\mathbb{N}\ \varphi_{n+1}(z)=o(\varphi_n(z))$ as $z\to z_0$

Asymptotic Expansion: $f(z)\sim\sum_{n=0}^{M}a_n\varphi_n(z)$ as $z\to z_0$ to M if $\phi_n(z)$ **is asy. seq.** and

$f(z)-\sum_{n=0}^{N}a_n\varphi_n(z)=o(\varphi_N(z))=\mathcal{O}(\varphi_{N+1}(z))$, $\forall N\le M$ as $z\to z_0$

$f(x,\varepsilon)\sim\sum_{n=0}^{\infty}a_n(x)\varepsilon^n$ uniformly in $x\in I$ as $\varepsilon\to 0$ if ε 与 x 无关(i.e. $\sup_{x\in I}|f(x,\varepsilon)-\sum_{n=0}^{N}a_n(x)\varepsilon^n|=o(\varepsilon^N)$)

Singular|Regular - Perturbation Method: **a.** Let $\varepsilon=0$ solve $f(x,0)=0$.

b.i. 如果 f 结构变了(阶数降低), Let $x=\varepsilon^{-\alpha}X$, PDB 找 α **c.** 假设 $X=a_0+a_1\varepsilon^{\beta}+a_2\varepsilon^{2\beta}+\cdots$ 代入求解

b.ii. 如果 f 结构不变/变, 但有重根 x_0, Let $x=x_0+\varepsilon^{\alpha}X$, PDB 找 α. $\uparrow\beta$ 看化简后的 X 方程得出

Principle of dominant balance (PDB): For Singular Perturbation $p(x,\varepsilon)=0$, Let $x=\varepsilon^{\alpha}X$, 代入选择 α:
对于 ε 来说 **a.** 至少有两项阶数相同且最低; **b.** 其他项阶数均更高. (即可忽略)

2 Integral By Parts, Laplace's Method & Method of Steepest Descents

Taylor: $f(x)=\sum_{n=0}^{N-1}\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n+\frac{f^{(N)}(\xi)}{N!}(x-x_0)^N\Rightarrow f(x)\sim\sum_{n=0}^{\infty}\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$ as $x\to x_0$

Exp. Integral: $E_1(x)=\int_x^{\infty}\frac{e^{-t}}{t}dt$, $(x\to 0$ 发散$)$ $E_1(x)\sim e^{-x}\sum_{n=0}^{\infty}(-1)^n\frac{n!}{x^{n+1}}, x\to\infty$

Gamma Func: $\Gamma(x)=\int_0^{\infty}t^{x-1}e^{-t}dt$ $\Gamma(x+1)=x\Gamma(x)$ and $\Gamma(n)=(n-1)!$ $\Gamma(\frac{1}{2})=\sqrt{\pi}$

Incomplete Gamma Func: **Lower**: $\gamma(a,x)=\int_0^{x}t^{a-1}e^{-t}dt$ **Upper**: $\Gamma(a,x)=\int_x^{\infty}t^{a-1}e^{-t}dt$

- $\gamma(a,x)=\Gamma(a)-\Gamma(a,x)$ $\Gamma(a,x)\sim e^{-x}x^{a-1}\sum_{n=0}^{\infty}\frac{\Gamma(a)}{\Gamma(a-n)}x^{-n}, x\to\infty$

Laplace's Method: For $I(x)=\int_a^b e^{xf(t)}g(t)dt$ as $x\to\infty$

1. If $f(t)$ has unique max at $t=t_0\in(a,b)\Rightarrow I\sim\int_{\infty}^{\infty}e^{x[f(t_0)+\frac{f''(t_0)}{2}(t-t_0)^2+\cdots]}g(t_0)dt$
   (留 n 项非常数展开为 n term approximation) (ps: max $\leftrightarrow f'(t_0)=0, f''(t_0)<0$)

   Useful Tool: Now: $I\sim e^{xf(t_0)}\int_{\infty}^{\infty}e^{x[\frac{f''(t_0)}{2}(t-t_0)^2+\cdots]}g(t_0)dt$ $\Rightarrow$ Let $\tau^2=-[\frac{1}{2}f''(t_0)(t-t_0)^2+\cdots]$
   Assume $t=\tau+b\tau^2+c\tau^3+\cdots$ as $\tau\to 0$ $\Rightarrow$ 代入换元式子求系数 $\Rightarrow$ $dt=(1+2b\tau+3c\tau^2+\cdots)d\tau$
2. **Exact Substitution**: 由 1 的基础, 得到 $f(t)$ 在 t_0 处的展开式, 换元令 $f(t)$ 就等于展开式前几项求解.
3. **If $g=g(t,x)$**: $e^{xf(t)}g(t,x)=e^{xf(t)+\ln(g(t,x))}$, 看 $xf(t)+\ln(g(t,x))$ 的最大值, 类似 1 处理.
   - If $t_0=f(x)$, 换元令 $t=f(x)s$.

Method of Steepest Descents: For $I(x)=\int_C e^{xf(z)}g(z)dz$ as $x\to\infty$. f 容易避开 **singular point**

If z_0 is saddle point: $f'(z_0)=0, f''(z_0)\neq 0$ $f(z)$ Taylor at z_0: $f(z)=f(z_0)+\frac{f''(z_0)}{2}(z-z_0)^2+\cdots$

1. Let $z=z_0+e^{i\theta}y$ (θ 选让 $f''(z_0)e^{2i\theta}$ 为负实数) $\to I(x)\sim\int_{-\infty}^{\infty}e^{x[f(z_0)-\frac{|f''(z_0)|}{2}y^2+\cdots]}g(z_0)e^{i\theta}dy$
2. If f 不容易避开 singular point: **a.** $\text{Im}(f(z))=\text{Im}(f(z_0))$ and $z_0\in C_0$ to find steepest descent path C_0
3. **b.** 画图看哪些 singular points 在 $C\to C_0$ 变形路径上, 计算 residues. **c.** $I=1$ 部分 + $2\pi i\sum$ residues.
4. **More than one saddle point**: i. 如果通过积分区域可以舍弃, 则舍弃 ii. saddle points: z_k, 取能让 $\text{Re}(f(z_k))$ 最小的.
   iii. 如果多个 saddle points 都满足 $\text{Re}(f(z_k))$ 最小, 则都要计算后叠加.
5. **Exact Substitution**: 由 1 的基础, 得到 $f(z)$ 在 z_0 处的展开式, 换元令 $f(z)$ 就等于展开式前几项求解.
6. **Remark**: 类似 Laplace 3 一样, 可以对 $xf(x)$ 一个整体处理, 必要时 $z_0=f(x)\to$ 换元 $z=f(x)s$.

Residue: If z_0 pole of order m of f, $\Rightarrow \text{Res}(f,z_0)=\frac{1}{(m-1)!}\lim_{z\to z_0}\frac{d^{m-1}}{dz^{m-1}}[(z-z_0)^m f(z)]$

Method of Stationary Phase: $\cos(xf(\theta))\to\text{Re}(e^{ixf(\theta)})$ 回到 Steepest Descents, 同理 $\sin(xf(\theta))$

3 Perturbation Methods in ODEs

Regular Perturbation: For $L(y,\varepsilon)=0$, let $y(x,\varepsilon)=y_0(x)+\varepsilon y_1(x)+\varepsilon^2y_2(x)+\cdots$
$\Rightarrow$ **a.** 代入 $\mathcal{O}(\varepsilon^n)$ $\to$ **b.** Boundary/Initial Condition 求解 $y_n(x)$ $\to$ **a.**

Singular Perturbation: e.g. $\varepsilon y''+p(x)y'+q(x)y=0,\ y(a)=A\ \ y(b)=B,\ a<b$, 大致步骤:

1. Assume boundary layer at $x=a|b$. Let $x=a+\delta(\varepsilon)X$ or $x=b-\delta(\varepsilon)X$
2. Assume inner $Y(X)=y(x)$, chain rule 求导代入原 ODE $\to$ Domain balance 找出 $\delta(\varepsilon)=\varepsilon^{\alpha}$
   Recall PDB: **a.** 至少有两项阶数相同且最低; **b.** 其他项阶数均更高.
3. Assume $Y=Y_0(X)+\varepsilon^{\beta}Y_1(X)+\varepsilon^{2\beta}Y_2(X)+\cdots$ 代入求解 $\to$ 边界/初始条件求解 $Y_0(X)$
4. **Matching**: Consider $X\to\infty|x\to a|b$, If limits not exist (除非 X) $\to$ boundary layer 在另一个端点 $\to$ 1
   If limits exist, boundary layer 正确 $\to$ 渐进 outer (x), 算 inner 极限 (X); 算不了保留 $X\to x$ 和 X 的关系代入 $\to$ 比较得系数.

Remark 1: Outer solution 只能用在 away from boundary layer 的区域 (包括边界点).

Remark 2: Inner solution 只能用在 boundary layer 附近的区域 (包括边界点).

Useful Lemma: For $\varepsilon y''+p(x)y'+q(x)y=0,\ y(a)=A\ \ y(b)=B,\ a<b, p(x)\neq 0$ in $[a,b]$ then
Has boundary layer at $x=a$ if $p>0$; Has boundary layer at $x=b$ if $p<0$.

4 Appendix

x, x^2, x^3 **as** $x\to 0$, x^{-1}, x^{-2}, x^{-3} **as** $x\to$ **inf**

4.1 Integral

$y'+Py=Q$: $\mu y=\int Q(x)\mu dx+C$, $\mu(x)=e^{\int P(x)dx}$

Gaussian Integral: $\int_{-\infty}^{\infty}e^{-ax^2}dx=\frac{\sqrt{\pi}}{\sqrt{a}}$ $\int_0^{\infty}e^{-xt^n}t^m=\frac{1}{n}\Gamma(\frac{m+1}{n})x^{-\frac{m+1}{n}}$

Integral: $\int\tan x=-\ln|\cos x|=\ln|\sec x|$ $\int\cot x=\ln|\sin x|=-\ln|\csc x|$

Techniques: **a.** e.g. $\int e^{at}\cos(bt)dt=\text{Re}(\int e^{(a+ib)t}dt)$

4.2 Expansion

Binomial Expansion: $(1+x)^{\alpha}\sim 1+\alpha x+\frac{\alpha(\alpha-1)}{2!}x^2+\cdots$ as $x\to 0$

Taylor|$x\to 0$: $f(x+h)\sim f(x)+f'(x)h+\frac{f''(x)}{2!}h^2+\cdots$ as $h\to 0$

$\sin(x)\sim x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots$ as $x\to 0$ $e^x\sim 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots$ as $x\to 0$

$\cos(x)\sim 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$ as $x\to 0$ $\ln(1+x)\sim x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots$ as $x\to 0$

$\sinh(x)\sim x+\frac{x^3}{3!}+\frac{x^5}{5!}+\cdots$ as $x\to 0$ $\cosh(x)\sim 1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots$ as $x\to 0$

$\tan(x)\sim x+\frac{x^3}{3}+\frac{2x^5}{15}+\cdots$ as $x\to 0$ $\cot(x)\sim\frac{1}{x}-\frac{x}{3}-\frac{x^3}{45}-\cdots$ as $x\to 0$

For $x\to\infty$: $\arctan(x)\sim\frac{\pi}{2}-\frac{1}{x}+\frac{1}{3x^3}-\cdots$ as $x\to\infty$

$\ln(a+x)=\ln x+\ln(1+\frac{a}{x})\sim\ln x+\frac{a}{x}-\frac{a^2}{2x^2}+\cdots$ as $x\to\infty$

$\frac{1}{a+x}=\frac{1}{x}\cdot\frac{1}{1+\frac{a}{x}}\sim\frac{1}{x}-\frac{a}{x^2}+\frac{a^2}{x^3}-\cdots$ as $x\to\infty$

sin|cos: $a\sin(x)+b\cos(x)=\sqrt{a^2+b^2}\sin(x+\varphi)$ $a\cos(x)+b\sin(x)=\sqrt{a^2+b^2}\cos(x-\varphi)$, φ 过 (a,b).

- $\sin(x)+\sin(y)=2\sin(\frac{x+y}{2})\cos(\frac{x-y}{2})$ | $\sin(x)\cos(y)=\frac{1}{2}[\sin(x+y)+\sin(x-y)]$ s+s=sc,s-s=cs,c+c=cc,s-s=-ss

Inequalities: $\ln(x+1)<x$, $e^x>x+1$, $(1+x)^a<e^{ax}$

4.3 Other

Nabla: $\nabla^2u=u_{xx}+u_{yy}+u_{zz}+\cdots$ $\nabla\cdot\mathbf{F}=\frac{\partial F_x}{\partial x}+\frac{\partial F_y}{\partial y}+\frac{\partial F_z}{\partial z}$